IFFO ELSANDE PRATAMA PUTRA

AI Engineer | ML Engineer | Data Scientist | Data Analyst | Data Engineer

Surabaya, Indonesia • +62 857-0440-0566 • elsandeputra890@gmail.com

LinkedIn • GitHub • Portfolio

PROFILE SUMMARY

Informatics Engineering Graduate from Universitas Negeri Surabaya with a GPA of 3.85/4.00. Experienced in machine learning, deep learning, and data analysis through IBM AI program, freelance data projects, and reinforcement learning-based undergraduate thesis research. Skilled in Python, TensorFlow, PyTorch, Scikit-Learn, and data science workflows, including data preprocessing, model development, performance evaluation, and visualization. Able to contribute to AI and data-driven projects by applying analytical thinking and machine learning approaches to real-world problems.

EDUCATION

Bachelor of Informatics Engineering (S1)

Aug 2021 – Jan 2026

Universitas Negeri Surabaya (UNESA) | Surabaya, East Java

- GPA: 3.85 / 4.00 — Graduated with Distinction
- Thesis: DoS Attack Classification on Blockchain Networks Using Proximal Policy Optimization (PPO) Algorithm — a study in reinforcement learning applied to network security
- Relevant Coursework: Data Science, Machine Learning, Deep Learning, Algorithms & Data Structures, Database Systems, Computer Networks, Cybersecurity

PROFESSIONAL EXPERIENCE

AI Engineer Intern

Aug 2026 – Present

PT. Platinum Ceramics Industry | Surabaya, East Java

- Developed an AI-powered Document Validation system to automate document verification and information extraction from invoices, purchase orders, delivery orders, tax invoices, receipts, and other business documents.
- Built an end-to-end Document AI pipeline combining OCR/Vision-Language Models (VLMs), document preprocessing, information extraction, and structured JSON generation.
- Implemented rule-based validation and document matching mechanisms to identify inconsistencies across extracted fields and related documents.
- Developed a web application to support document validation workflows and present analysis results in a structured format.
- Evaluated document extraction performance and analyzed failure cases to improve system reliability.

Machine Learning Engineer (Freelance)

Jul 2024 – Present

Cons Code | Surabaya, East Java

- Perform data cleaning, preprocessing, and transformation tasks using Pandas and NumPy, producing structured datasets ready for analysis or model training.
- Develop simple Python-based classification and prediction scripts based on client requirements, contributing to basic data-driven decision-making workflows.
- Supported data analysis tasks and helped generate summary reports from processed datasets, presenting findings in formats accessible to non-technical stakeholders.
- Explored and applied model evaluation metrics including accuracy, precision, recall, F1-score, and AUC-ROC to assess solution quality and report results to clients.

AI Engineering Intern – IBM Advanced AI Program

Feb 2024 – Jun 2024

PT Kinema Systrans Multimedia (Independent Study) | Batam, Riau

- Participated in IBM's structured Advanced AI Academy program covering generative AI concepts, AI governance frameworks, and enterprise data management using Watsonx.data platform.
- Practiced building, training, and testing deep learning classification models using TensorFlow and PyTorch through hands-on exercises in the program curriculum.
- Studied LLM fundamentals, basic automation pipeline concepts, and AI deployment considerations as part of the program's applied AI modules.
- Learned and applied cybersecurity and data management best practices relevant to responsible AI system development in enterprise settings.

Web Developer Intern *Oct 2024 – Jan 2025*

PT Radnet Digital Indonesia | Surabaya, East Java

- Building company profile and e-commerce websites for MSME clients using the Django framework, covering the full cycle from design to deployment.
- Developed responsive front-end pages including product catalog, contact forms, and a basic CMS, integrated with Django back-end using template rendering and REST API.
- Set up and maintained Linux-based server environments for web deployment, learning practical server management and basic security configuration in a real working setup.
- Participated as a speaker in a webinar on web development fundamentals attended by 20+ participants, practicing the ability to explain technical concepts to a broader audience.
- Gained first-hand experience working in a professional team environment with structured workflows, real client deadlines, and collaborative code reviews.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, Java, C++
AI / ML	TensorFlow, PyTorch, Scikit-Learn, Keras, Hugging Face, OpenCV
Data Science	Pandas, NumPy, Matplotlib, Seaborn, Scipy, Jupyter Notebook
BI & Analytics	Power BI, Tableau, SQL Server, Watsonx.data
MLOps	Docker, GitHub Actions, CI/CD, Git, FastAPI, Streamlit, Roboflow
Cloud	AWS, GCP, Azure, IBM Cloud
Domains	Computer Vision, NLP, Predictive Analytics, Reinforcement Learning, Cybersecurity AI
Soft Skills	Fast Learner, Analytical Thinking, Research-Oriented, Communication, Teamwork

PROJECTS

DoS Attack Classification on Blockchain Networks using PPO (Reinforcement Learning)

Reinforcement Learning • PPO • Python • Network Security • Cybersecurity AI

Undergraduate thesis project: designed, implemented, and evaluated a Proximal Policy Optimization (PPO)-based reinforcement learning model to detect and classify Denial-of-Service attacks on blockchain networks. Included a literature review, multiple model experiments, and performance analysis — submitted as an original academic research contribution.

Disease Detection in Tomato Plants

TensorFlow • TensorFlow Hub • ResNet-v2 • Transfer Learning • Image Classification

Academic project: developed a multi-class image classification model using a pretrained ResNet-v2 architecture via TensorFlow Hub to identify diseases in tomato plant images across multiple categories. Practical exercise in transfer learning, model fine-tuning, and image preprocessing pipelines.

Used Goods Classification (Computer Vision + MLOps)

Roboflow • Data Augmentation • Image Preprocessing • Object Detection • API Deployment

Personal project: manually collected and labeled image datasets from multiple online sources, applied preprocessing and data augmentation using Roboflow, trained an object classification model, and deployed it via the Roboflow API for real-time inference demonstration.

Diabetes Classification System

Random Forest • MLP • Scikit-Learn • Python • AUC-ROC • Predictive Analytics

Academic project: implemented and benchmarked Random Forest and MLP classifiers on a patient diabetes dataset. Evaluated model performance using accuracy, precision, recall, F1-score, and AUC-ROC — a practical end-to-end exercise in supervised learning, feature analysis, and model comparison.

CERTIFICATIONS & TRAINING

IBM Advanced AI Academy *Feb – Jun 2024*

PT Kinema Systrans Multimedia | Batam, Riau

- Topics covered: Generative AI, AI Governance, Watsonx.data, Cybersecurity in AI, and Deep Learning Foundations

Data Science & Machine Learning Fundamentals *Jan 2024*

Dicoding Indonesia | Bandung, West Java

- Topics covered: Data preprocessing, supervised & unsupervised learning, model evaluation, and model deployment